Worksheet — Buoyancy Simulation

Use with *Simulation - Buoyancy.html* · goes with Station 4 (Archimedes' principle)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/08-buoyancy** in a browser (offline: *Simulation - Buoyancy.html*). Choose the object, its volume and the liquid, then lower it with d (▲ ▼). The overflow runs into the beaker on the balance. ✋ **Let go** releases the object.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab, Station 4 (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}, ρ_{water} = 1000 kg m^{-3}. 1 cm^3 = 1 × 10^{-6} m^3.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. As the cylinder goes into the water, what happens to the newton meter reading?			
2. Fully under water, you lower it further (not touching the bottom). Reading?			
3. Compare the buoyancy force with the weight of the water collected.			
4. You lower a pine block (ρ = 500 kg m^{-3}). What happens?			

5. Draw the free-body diagram of the cylinder hanging fully under water, then calculate the buoyancy force on the 63 cm^3 cylinder.

Free-body diagram and working

Part 2 — During the lab: Station 4

	Weight in air / N	Apparent weight in water / N	F_b / N	Displaced water / kg	its weight / N
Simulation					
Our measurement					

6. Does your data support Archimedes' principle? Give one reason why the two values in your own row are not exactly equal.

7. Why does a steel ship float while a steel nail sinks? Use your plasticine boat to explain.

Part 3 — Homework

8. Record at least 5 fully-submerged combinations (objects, volumes, liquids). Plot F_b against the weight of liquid displaced. Gradient = _____

Table of results

9. Mystery stone: weight in air = _____ N, in water = _____ N. Calculate its density.

Working

10. What fraction of a floating ice cube (ρ = 920 kg m^{-3}) is under water? What fraction of a cork (ρ = 240 kg m^{-3}) in saturated salt water (ρ = 1200 kg m^{-3})?

Working

11. A steel block (ρ = 7850 kg m^{-3}) is put in mercury (ρ = 13 600 kg m^{-3}). Does it float? Draw its free-body diagram and find the fraction submerged.

Working